Rhetoric on political manifesto for French legislative elections of 1993

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1 Introduction

Elections are one of the pillars of democracy, and the main instrument allowing people to make their voices heard. As the political candidates seek to make the most people adhere to their ideas and views, they must campaign to find this support. Political manifestos thus play a crucial role in elections as they reveal the strategies adopted by candidates, and by political parties. The Archelec corpus preserves those manifestos for important French election, particularly those that happened in the second half of the 20th century. this allows several studies to be caried, on the content of those manifestos, and on the way candidates tried to win over their audience.

The focus point of this analysis of political manifestos is the rhetoric used by candidates in these manifestos, and how they use language to rally voters. Especially, I am interested in how politician addresses their public, and they are mobilizing them to act. Often, in their speeches, or in our case in their manifestos, politicians will either uses sentences to describe the actions they are planning to carry on (or have carried on), or sentences to include the voters in their agenda. This is specifically this rethoric that I am trying to analyse. Indeed, one can wonder if certain usages are more common to certain parties, or to certain candidates. I am particularly focusing on the following question : Is there a gender difference in the ways of addressing the voters ? But I will also try several other features, such as the candidate's occupation and chere the election takes place, to understand more the data.

Natural Language Processing offers several useful tools to answer that question, by allowing a thorought exploration and analysis of a consequent corpus of political manifestos from the 1993 French general election. The base tool allowing this analysis is the "Part Of Speech" (POS) tagging technique, that attach tag to words, in our case depending on their syntaxical role. This tool will be used in order to single out pronouns usage and the following verbs. Other metrics, such as Textual Lexical Diversity, will also be used to extend the analysis. Finally, a fine-tuned CamemBERT model will be deployed to differentiate verbs carrying concrete actions, and more passive words.

2 State of the art

This study is partially inspired by [2]. In their article, they explore how politicians appeal to social group during electoral periods. While they mostly focuses on addresses to social group, a very interesting topic, I wondered how on political manifestos, before an election, politicians could not only addresses their electorate, but also mobilize their voters to rally behind them, either by addressing them personnally, to call them to act, or to underline their own actions. Moreover, the methodology used in this article reveals itself very interesting and robust, altough it will not be used in this study (as I will focus on a slightly different subject and perspective).

One of the first idea for this analysis, was to study the use of pronouns. Indeed, it gives an idea on how the speeches are constructed, and to whom they are directed to. For this task, the main tool that will be used are POS taggers. [3] describes how POS works. Especially, it highlights how important this tool is for several other NLP tools. This importance lead to the development of robust techniques used in POS tagging. The main point of relevance of POS tagging, is that most of the time it is not possible to simply rely on a dictionary to identify syntaxical functions. The techniques developped

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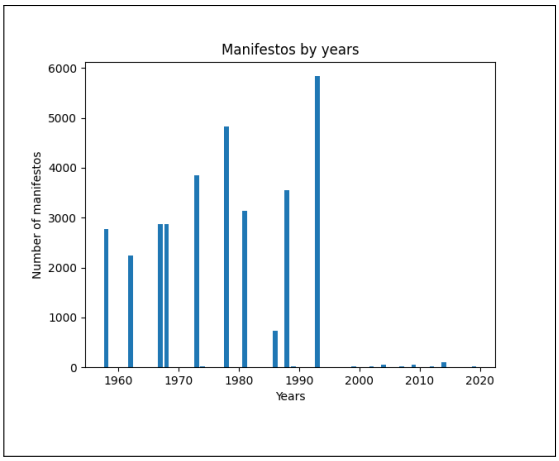

Figure 1: Number of manifestos by year, in the Archelec corpus. The 1973 legislative election is the most represented.

includes n-gram taggers (Hidden Markov Model, using n-1 words to predict the following), neural networks and decision tree taggers. All these techniques achieves very high level of accurracy, and POS tagging is a very reliable tool from which I will be able to easily interpret results, even in a French corpus.

In this study, I also aim at differenciate verbs use in the manifestos. Indeed, I want to look into how "active-leaning" verbs used by the candidates are, whether about their own action, or a call to action. In order to achieve that, I took inspiration from [1] who used classification tools to separate opinions from news in French press article. In their study, they compare CamemBERT and another machine learning model. Models based on the *transformer* architecture, such as CamemBERT, have known a significant development, and improvement of their performances. In their article, they successfully fine-tune this model on subjective classification task, something that had not really been done before. Despite the important computational ressources needed to apply this model and its lack of explainability, it appeared as the most relevant in my case.

3 The Data : the Archelec corpus

The Archelec corpus contains manifestos of French legislative and presidential elections of the second half of the 20th century. The following Figure 1 depicts the repartition of manifestos available for each year. The legislative election of 1993 is the most represented, and I will focus solely on this election for the remaining of the study.

This study aims to focus on gender differences in the rhetoric used in those manifestos, it is thus relevant to check the gender repartition of the main candidates on each manifestos. Because it is a legislative election, each manifestos presents a list of candidates, out of whom one is heading the list. It is this candidate that I choose to hold as representative of the whole list in this study.

Figure 2 shows how unbalanced the gender repartition of candidates (heading the list) is. However, statistical studies will still be able to be carried out, using undersampling methods.

This data is well structured, but needs some formating. The maint change needed, is the cleaning of the texts themselves. There are, indeed, many phrases and expressions that do not add any value to those texts (such as "ELECTION FRANCAISES"). All this content is not needed for this analysis, and I thus used Regular Expression to remove all of it (especially the name of the other candidates in the same list, and the circonscription in which they run). This formating is not perfect, but it still allows to clean a lot of undesired text, that would have hindered the study.

In order to explore a little bit the data, I studied the frequency of the use of the pronouns "Je", "Vous" and "Nous" in the manifestos, for every 1000 words, and compared this usage depending on the gender. The results are shown in Figure 3.

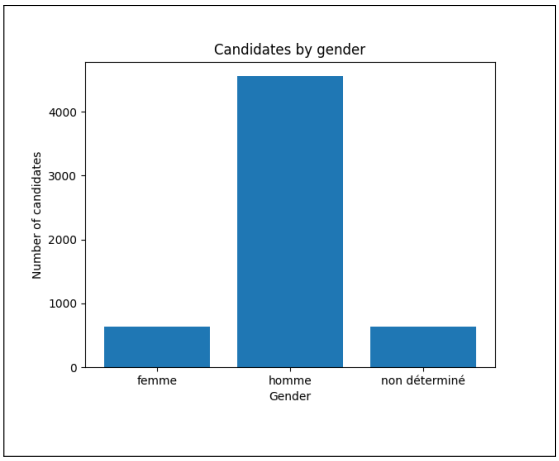

Figure 2: Gender repartition of the main candidate on each manifestos. There is an overwhelmingly majority of male candidates.

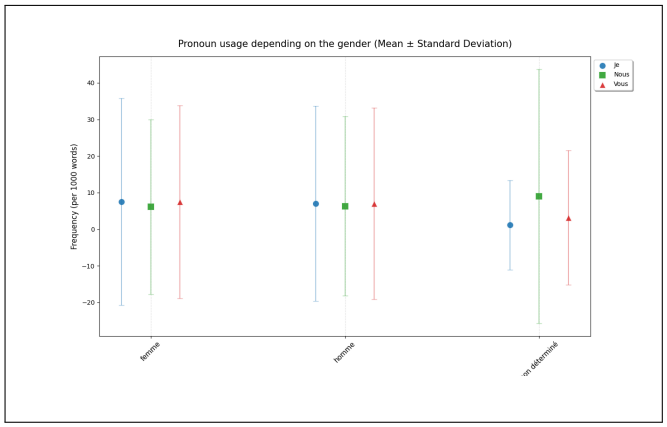

Figure 3: Pronouns usage depending on the gender of the candidate. It is overall evenly distributed.

The usage of those three pronouns are very evenly distributed between genders. The only slight divergence are from the "Undetermined" group, which may not be very relevant in this analysis, as it cannot be studied in any way.

However, this analysis remains relevant. Indeed, even thought the use of personnal pronouns seems to be constant between gender, what will be very interesting is to analyse the verbs following those pronouns, and the constant use of pronouns between gender allows for a legitimate comparison.

To finish this primary exploration of the data, I evaluated the lexical diversity of each text. For that, I used the Type Token Ration (TTR) method. They are defined as the ratio of unique token (here words) on the total number of token. In our case, I calculated fro each text the number of unique words, divided by the total number of words. I then compared the results depending on the gender of the candidate, and depending on the department where the candidate runs. Figure 4 show the results (to help readibility, the results by departments are presented in Figure 7 placed in the appendix). Lexical diversity is fairly similar throughtout all modalities, both for gender, and localisation (despite the numerous departments).

I also looked throught the difference depending on the candidate's occupation. Table 1 displays the occupation with the most (and least) lexically diverse manifestos. Different occupations seem to have more of an impact on the lexical diversity of manifestos as the most lexical diverse occupation is almost twice as diverse as the occupation with the least diverse manifestos. However, consider the number of possible modality for this variable, it will be hard to use it in this analysis.

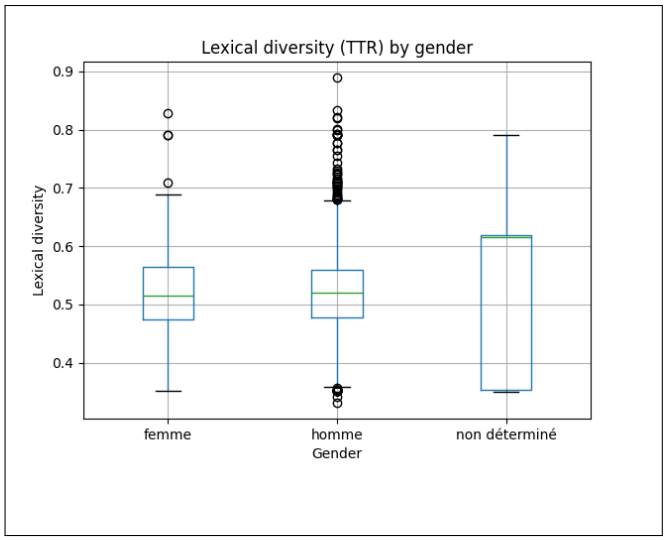

Figure 4: Lexical diversity depending on the gender. It is overall evenly distributed.

Table 1: Occupation with the most, and least, lexical diverse manifestos. It is less balanced than other features.

Lexical Diversity		
Rank	TTR	Occupation
1204	0.350837	Aide-soignant
1201	0.351104	Electro-technicien automobile
1201	0.351104	Ouvrier monteur
1201	0.351104	Employée hôpitaux
1200	0.351774	Chef d'entreprise; Profession libérale
...	...	...
5	0.658986	Responsable financier
4	0.664430	Commerçante alimentation
3	0.677419	Responsable centre télécom
2	0.708333	Commerçant restaurateur
1	0.764706	Radio

Overall, what the data can show already us with those fairly simple NLP tools (POS tagging and Type Token Ratio) is that the rhetoric used on manifestos (as far as pronouns usage and lexical diversity go) are fairly even between gender, and department, however they differ more importantly depending on the occupation of the candidate.

4 Type of verbs used : fine-tuning CamemBERT

To push the analysis further, I wanted to focus on the content of those manifestos. Especially, I wanted to see how the candidates expressed themselves, and how active they were. In order to do that, I chose to focus on the verbs they used in each sentence where they use a "je", "nous" or "vous" pronouns, and I processed to classify them in two classes : "concrete" verbs, meaning verbs referencing a concret action (either done or planned), and "contextual" verbs, meaning verbs more passive and giving context, often in a superficial way.

In order to do that, I chose to fine-tune a Transformers-model CamemBERT. I annotated, on a 500 phrases sample, all the contextual verbs and all the concrete verbs. I then use this sample to train the model, to apply it on the whole corpus of manifestos from the 1993 French legislative elections. After this, I was able to analyse these results, to identify the difference in rhetoric.

Table 2: Departments with the most, and least, usage of contextual and concrete verbs.

Rank Contextual	Department	Rank Concrete	Department
1	Charente	1	Charente
2	Corse du Sud	2	Alliers
3	Bas-Rhin	3	Haute-Savoie
...	...	...	...
94	Eure-et-Loir	94	Loire-Atlantique
95	Ariège	95	Ariège
96	Mayenne	95	Ille-et-Vilaine

As I have underlined earlier, there is a clear imbalance between male and female candidates. In order to reduce bias on the analysis, I undersampled the male lead manifestos, and I conducted my analysis on this sample, for the gender modality.

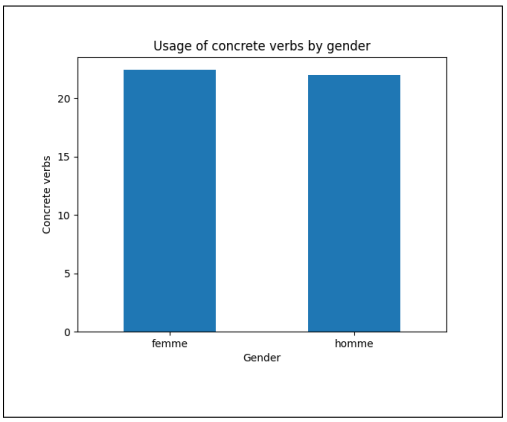

Figure 5: Concrete verbs use

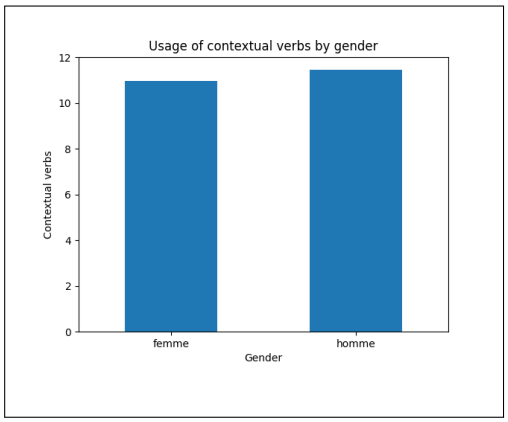

Figure 6: Contextual verbs use

Figure4 displays results for both concrete verbs and contextual verbs use depending on the gender. It appears that men tend to use slighty more contextual words than women, whereas women use a little bit more concrete verbs (the difference is slim, but a little bit less important than that on the contextual verbs use). This result, altough not very conclusive is very interesting because it disrupts the classical image that was held of women in politics during a long time. It could well be that this use of a more active vocabulary by the women is a result of an opposition to the way women in politics were seen previously (as, maybe less fit for the job, less active..), resulting in a kind of overcompensation from the femal politicians.

I then conducted the same analysis for the departement of the election. As for the difference between the departments, the results are shown in the Appendix (Figure8 and Figure9), for readibility reasons. I exctracted the departments with the most, and least usage of both concrete and contextual verbs in the Table2. What we can see with these results, is that there is indeed different use in concrete and contextual verbs, however those results may lack other explanations. Charente, for example, is the department where manifestos are both using the most concrete verbs, but also the most contextual verbs. Thus, it is probably juste a department where the manifestos contained, overall, more verbs than other department, independantly of the type of these verbs. The same can be said about Ariège, which inversely, is a department where manifestos contains less concrete and contextual verbs. Even thought this results do not shed lght on the problem, it shows a diversity of usage, confirming the relevance of the model.

5 Conclusion

To conclude, this study focused on the difference in rhetoric on political manifestos, for the 1993 legislative election in France. Several features were taken into consideration, most prominently

gender, but also the department in which the election took place and the occupation of the lead candidate. Using several NLP tools, I was able to compare and define differences (or similarities) in rhetorics. Using POS tagging, I showed that the use of the pronouns "je", "nous", "vous" was fairly similar accross all gender, denoting a constant desire to involve both themselves and their voters in their manifestos. Similarly, the study of lexical diversity (through type token ratio) showed similar results, where gender and the localisation had no impact, whereas the candidate occupation seemed to have more of an impact. To round up this analysis, the fine-tuning of a CamemBERT model underlined a very small difference in the use of more concrete verbs depending on the gender, that could find explanations in the history of women in politics, whereas the localisation had more of an impact, as different departments had very different scores. Even thought this analysis underlined several aspects of this corpus, and especially a lack of clearly defined gender-specific rhetoric, more thorought analysis could be carried out to understand more what could influence rhetoric in political manifestos.

References

[1] Louis Escouflaire, Antonin Descampe, and Cédrick Fairon. Automated text classification of opinion vs. news french press articles. a comparison of transformer and feature-based approaches. *Language Communication*, 99:129–140, 2024.

[2] Hauke Licht and Ronja Sczepanski. Detecting group mentions in political rhetoric a supervised learning approach. *British Journal of Political Science*, 55:e119, 2025.

[3] Giorgos S. Orphanos, Dimitris Kalles, Thanasis Papagelis, and Dimitris N. Christodoulakis. Decision trees and nlp: A case study in pos tagging. 2009.

Appendix

The following figures are placed here for readibility reasons.

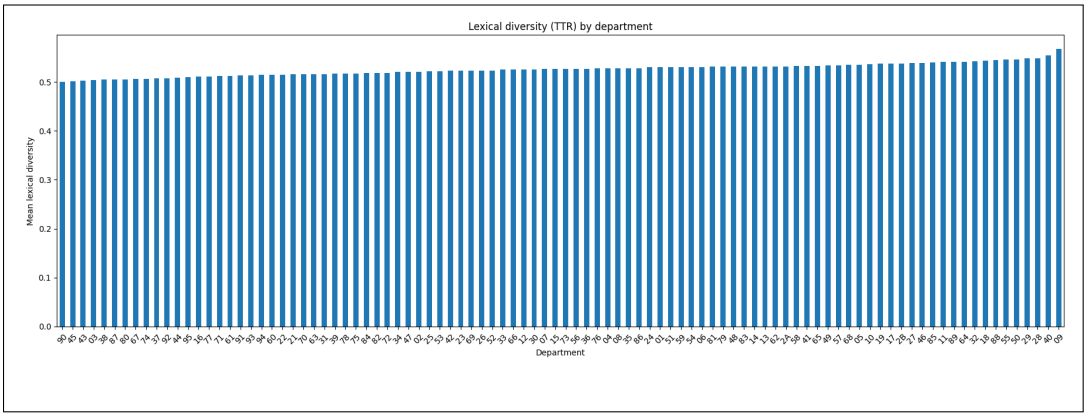

Figure 7: Lexical diversity depending on the department of the election. It is overall evenly distributed.

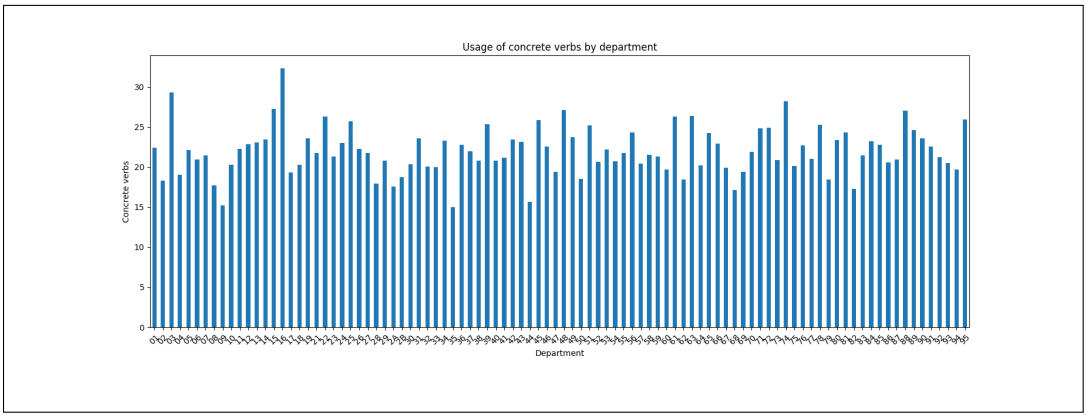

Figure 8: Concrete verbs use dependig on the department of the election.

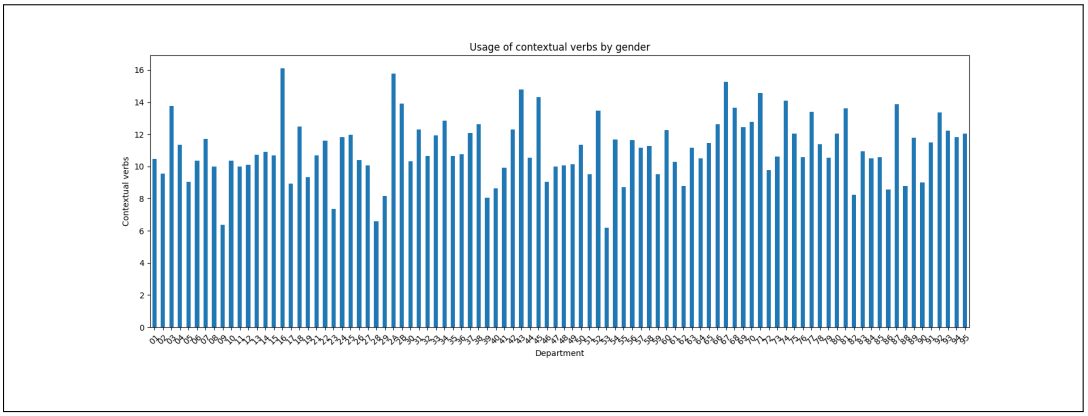

Figure 9: Contextual verbs use dependig on the department of the election.